Visualizing Patient Flows in Brazil's Unified Health System: An Interactive Map of SIH/SUS Hospitalizations

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Abstract— Brazil's Unified Health System (SUS) records every publicly funded hospital admission in the SIH/SUS dataset, a general-purpose source that also stores the patient's municipality of residence and the municipality where the admission took place. Together these two fields describe how patients move across the country to be hospitalized, yet the official tabular access hides this movement and can conceal where a region depends on outside care. We present an interactive flow map that draws these movements as origin-destination arcs over a map of Brazil. The view is organized in three levels, country, state, and city, and supports filtering by medical specialty, year, and month, along with side panels that rank attracting and dependent municipalities and compare resident and incoming admissions across the year. Using 183 million admissions from 2008 to 2023, we present case studies that illustrate how the tool is designed to surface care gaps, regional hubs, and seasonal patterns that tabular access tends to obscure. This paper describes the design and the system; an evaluation with users remains future work.

Index Terms—Geographic visualization, flow maps, origin-destination data, public health, interactive visualization.

1 Introduction

Brazil's Unified Health System (SUS) guarantees every resident the right to publicly funded hospital care. In practice, hospitals and medical specialties are distributed unevenly across the territory, so a large number of patients are admitted far from where they live. Knowing where these patients go, and for which specialties, is valuable for planning hospital capacity and for identifying regions that depend on care produced elsewhere.

The Hospital Information System of SUS (SIH/SUS), published by DATASUS, records every publicly funded admission. It is a general-purpose dataset, since each record carries more than one hundred fields, among them diagnoses, procedures, costs, length of stay, intensive care use, and patient demographics. The same record also stores the patient's municipality of residence and the municipality of the hospital, so a single admission already describes a movement between two places.

This work sets the clinical and financial fields aside and concentrates on that movement. Origin and destination together expose problems of access that the other fields do not show directly. The flow of patients can point to gaps in care, bottlenecks at regional hubs, and local overload.

Our extract holds 183 million admissions from 2008 to 2023. The official route to this public data is the DATASUS tabulation portal, which returns the records as configurable tables. A table is a weak instrument for this question. Reading movement among thousands of municipalities from rows and columns is impractical, and the aggregate counts a table reports can hide where a region depends on care produced elsewhere.

We present an interactive flow map of these hospitalization flows. Each arc links a patient's residence to the hospital municipality, and its width grows with the number of admissions. The view operates at three levels, from the whole country down to a state and then a single municipality, and it can be filtered by specialty, year, and month.

We do not claim to identify every gap or bottleneck in the system, nor do we evaluate the tool with users. This paper is a design and system description: we explain how the visualization is built to make these patterns easier to see than a table allows, and we present case studies that illustrate its use. A study with health managers is left for future work.

The design supports three themes of analysis that motivated the project. The first is gaps in care, where the residents of a region must travel for a given specialty, which can show a government where attention is needed. The second is hub cities that concentrate the care of a whole region, where the demand they absorb may signal overload or a bottleneck for planning. The third is seasonality: whether a monthly peak falls on residents or on incoming patients, and whether a city serves its own residents first or absorbs an influx from elsewhere. The remaining sections describe the data preparation and the design of the visualization, and then present a case study for each theme.

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2 Related Work

Several studies use SIH/SUS data to study where Brazilians are hospitalized. Oliveira et al. [4] mapped hospital care networks and found that most admissions still happen in the patient's own municipality, while the remaining flows point toward regional hubs. Rocha et al. [5] measured patient migration between health regions in São Paulo and classified each region as attracting or losing patients. Xavier et al. [8] used the origin and destination of admissions to build health regions and to locate care gaps. These works report indices, tables, and static maps, but they rarely let the reader explore the flows directly.

Information visualization has long studied how to draw origin-destination (OD) data. Drawing every flow as a straight line produces heavy occlusion, so much of the work looks for ways to reduce clutter. Wood et al. [7] replaced lines with small gridded maps (OD maps) to avoid crossing flows, while Jenny et al. [3] proposed design principles for line-based flow maps and Graser et al. [2] compared techniques for untangling OD flows in a GIS. For data that changes over time, Flowstrates [1] separates origins, destinations, and a temporal heatmap. We borrow the clutter-reduction goal from this line of work and apply it through three concrete choices: curved arcs in place of straight lines, aggregation by administrative level so that only a tractable number of arcs is drawn at once, and a direction filter that hides the arcs the user is not inspecting.

Interaction is what lets users work with large and multivariate data. Shneiderman's mantra of overview first, zoom and filter, then details on demand [6], together with the interaction categories of Yi et al. [9], describes the operations our tool depends on. The three-level drill-down and the specialty and time filters put these ideas to work on the hospitalization data.

General flow-map tools such as kepler.gl and flowmap.blue can plot OD data, but they are generic plotters that a user must configure for each dataset. Our work connects the two threads above and adds three features that a generic plotter does not provide on this data. First, a three-level drill-down bound to the Brazilian administrative hierarchy,

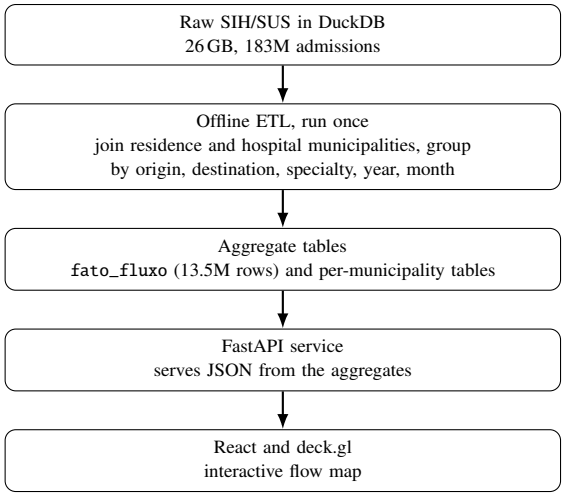


Fig. 1: Data pipeline. All aggregation runs once, offline, turning the 183 million row source table into small tables that keep the views interactive; the text reports measured query times.

from federative unit to municipality. Second, a precomputed aggregation that keeps the full 183 million record dataset interactive. Third, panels built for the access questions at hand: an attraction ranking and a resident-versus-incoming seasonal comparison.

3 Proposed Visualization

3.1 Data preparation

The raw SIH/SUS extract is too large to query while the user waits. It holds the admissions in a 26 GB table, and a single aggregation over one year already takes tens of seconds. To keep interaction fast, we move all of the heavy work to an offline step that runs once.

This step makes one pass over the fact table. It joins each admission to the municipality where the patient lives and to the municipality of the hospital, then groups the rows by origin, destination, specialty, year, and month. Municipality codes follow the six-digit IBGE scheme and resolve against a municipality table that carries latitude and longitude. The window from 2008 to 2023 covers 183,274,245 admissions. No record is dropped at this stage, since every admission carries a valid municipality code; only 19 of the 5,589 municipalities (0.3 percent) lack coordinates, so the few flows that touch them are not drawn on the map. The specialty used throughout is the bed-specialty field of SIH/SUS (ESPEC), whose coverage is uneven, a limitation we return to in the Conclusion.

The pass produces three aggregate tables, summarized in Figure 1. The flow table, with 13.5 million rows, keeps one row per origin, destination, specialty, year, and month, and feeds the arcs. Two per-municipality tables, one by year and one by year and month, store the admissions each place receives and generates; they feed the ranking, while the monthly resident and incoming series are read back from the flow table.

3.2 The flow map

We draw each hospitalization flow as a curved arc from the patient's residence to the hospital municipality. The width of the arc grows with the number of admissions, so the thickest arcs mark the busiest routes. Color marks the direction relative to the place in focus, with arcs that leave a region drawn in orange and arcs that arrive drawn in blue.

A choropleth sits behind the arcs and fills each state according to a metric chosen by the user. The tool offers five metrics, shown for the whole country in Figure 2. The default is the *attraction balance*: the admissions a place receives minus the admissions its own residents generate, on a diverging blue-to-red scale, where blue marks a net attractor and red a place that sends away more patients than it receives. The *export rate* is the share of a place's resident admissions that are treated in another municipality, on a yellow-to-red scale. Two metrics report absolute volume, the *admissions received* on a blue scale and the *admissions exported* on an orange scale. The fifth, a *per-capita rate*, divides the admissions received by the local population and multiplies by one thousand, on a purple scale; it draws on IBGE population figures and is available for 2008 to 2021, the years those figures cover. City nodes appear as circles sized by the number of admissions that pass through them, over a dark basemap that keeps the arcs and nodes distinct.

The arc colors and the choropleth scales are independent encodings, and the legend states which is active: orange and blue mark arc direction, while the choropleth uses its own per-metric scale and is darkened so the brighter arcs stay distinct.

3.3 Three levels of detail

The map follows the principle of overview first, then zoom and filter, then details on demand [6]. It opens at the national level, where the 27 federative units (26 states and the Federal District) are the nodes and the arcs show flows between them. Clicking a state opens the state level, where its municipalities become the nodes and the arcs show flows inside the state and toward other states. Clicking a city opens the city level, which shows every flow that touches that city across the country. Figure 3 shows the three levels for Rio Grande do Sul. A breadcrumb at the top records the path and lets the user step back, and the camera flies to each new selection.

3.4 Filters and animation

Filters narrow the data without leaving the map. The user can pick a medical specialty, a range of years, and a single month, and can change the metric used for the choropleth. A direction control limits the arcs to those leaving or arriving at the focus, which removes clutter when a city has many connections. A further option plays the twelve months in sequence, so seasonal change appears as motion on the map.

3.5 Analytical panels

Two side panels sit next to the map, both visible in Figure 3. The first ranks municipalities by the same attraction balance used for the choropleth. A toggle reverses the order between the places that attract the most patients and the places that most depend on others.

The second panel plots a monthly time series that separates resident admissions from incoming ones. This shows whether a seasonal peak comes from local patients or from people arriving from elsewhere. The interface is in Portuguese, where this panel labels the two series *Residentes* and *Migrantes*; we call them resident and incoming patients in this paper.

Selecting a row in either panel moves the map to that municipality, so a reading in a panel and the matching shape on the map stay linked.

4 Usage Scenarios

We now apply the tool to the three themes set out in the Introduction. These are usage scenarios rather than a systematic study. Each case below was chosen by hand to demonstrate a capability, not sampled by a defined criterion, so its figures describe the chosen places and do not generalize on their own. The scenarios use different year windows. The spatial snapshots use a recent complete year or a multi-year window that accumulates enough volume, with 2023 the most recent year. The seasonal analysis uses windows that end before 2023, because the final months of the latest year are still incomplete and would distort a monthly curve. As a sense of scale, across the full 2008 to 2023 extract 32 percent of the 183.3 million admissions (58.6 million) were recorded outside the patient's home municipality, counting an admission as exported when its residence and hospital municipalities differ. Both counts come from the same aggregated dataset described in Section 3, so cross-border flow is substantial and worth visualizing.

4.1 Gaps in care for a specialty

The first theme is the gap in care, where the residents of a region must travel for a given specialty. A municipality can be a hub for one specialty and dependent for another. Figure 4 shows Santa Maria, a hub

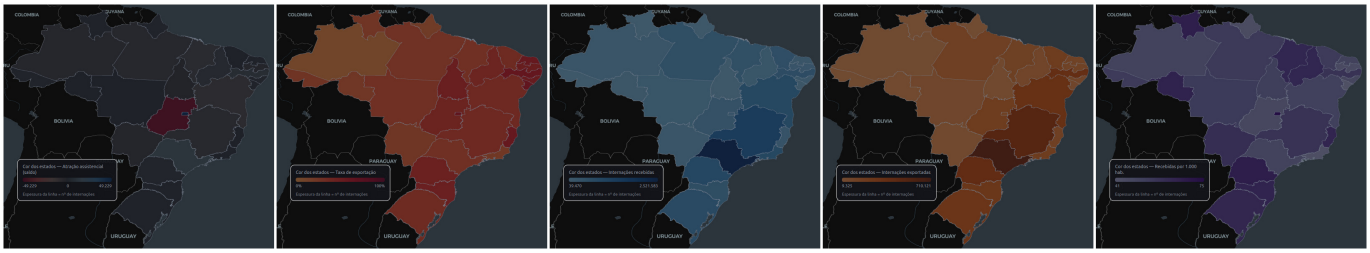


Fig. 2: The five choropleth metrics shown for Brazil at the national level, from left to right: the attraction balance, the export rate, the admissions received, the admissions exported, and the admissions received per 1,000 inhabitants. Each metric uses its own color scale, so the same map answers a different question.

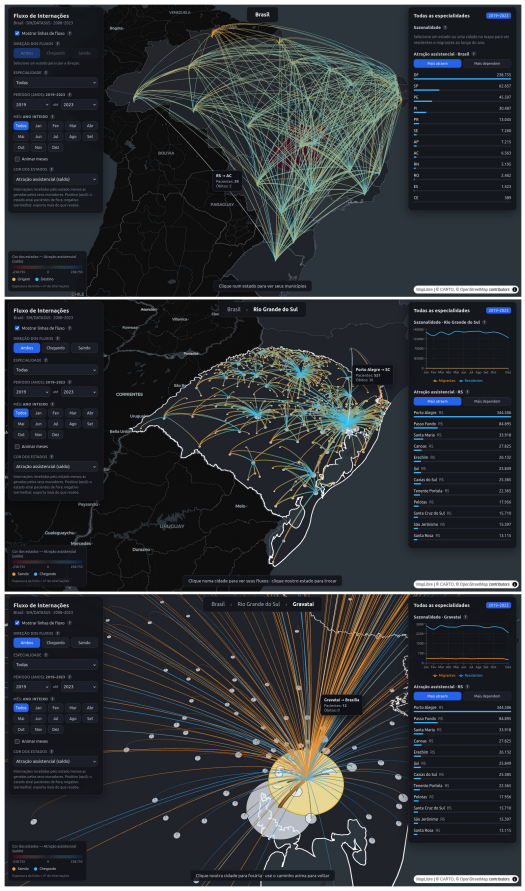


Fig. 3: The three levels of the flow map for Rio Grande do Sul. Top: the national level, with states as nodes and arcs between them. Middle: the state level, with municipalities as nodes. Bottom: the city level, showing every flow that touches the selected city.

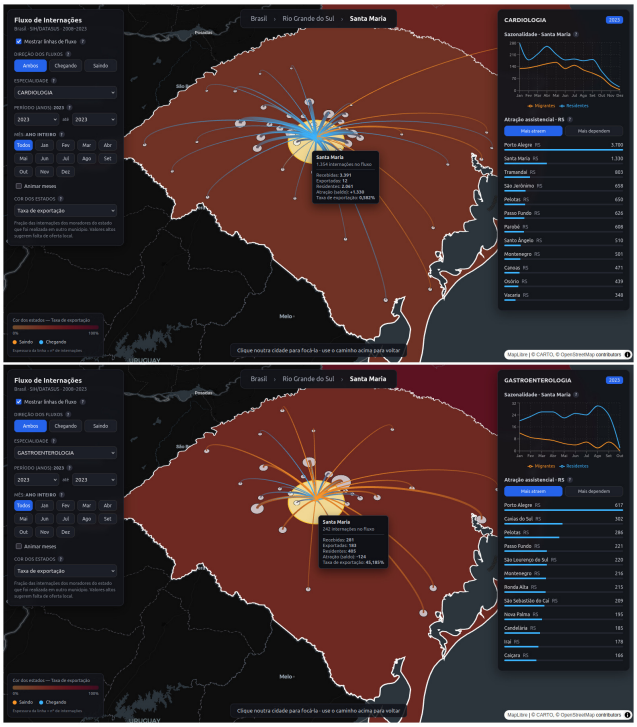


Fig. 4: Santa Maria (Rio Grande do Sul) under two specialties in 2023. Top: cardiology, where the city is a hub reached by incoming blue arcs (balance $+1{,}330$, export rate 0.582 percent). Bottom: gastroenterology, where its residents leave along orange arcs (balance -124, export rate 45.185 percent). Only the specialty filter changed between the two views.

in the interior of Rio Grande do Sul, under two specialties in 2023, the most recent complete year. For cardiology it received 3,391 admissions against 2,061 from its own residents, a positive balance of 1,330 and an export rate of 0.58 percent: only 12 of the 2,061 resident admissions were treated elsewhere. For gastroenterology it received 281 against 405 residents, a negative balance of 124 and an export rate of 45.18 percent (183 of 405).

Changing only the specialty filter turns Santa Maria from a destination reached by many incoming blue arcs into an origin whose residents leave along outgoing orange arcs. A pattern of this kind points a health authority to a specialty and a region that may need attention. Because the coverage of the specialty field is uneven, the reading is most reliable where a specialty has many admissions.

4.2 Hub cities and regional load

The second theme is the hub city that concentrates the care of a region, where the demand it absorbs may signal overload or a bottleneck. The ranking panel orders municipalities by their attraction balance. Figure 5 shows Rio Grande do Sul over 2019 to 2023, where Porto

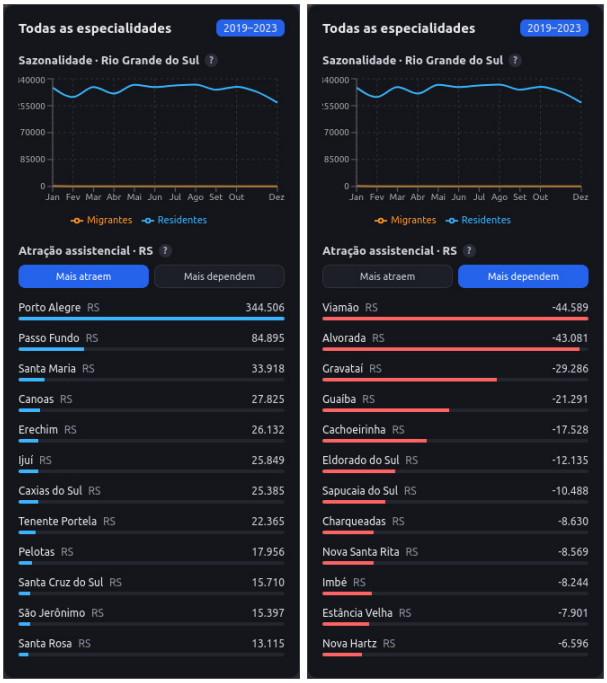


Fig. 5: The attraction ranking for Rio Grande do Sul, all specialties, 2019 to 2023. Left: the municipalities that attract the most patients, led by Porto Alegre. Right: the municipalities that most depend on care elsewhere, led by Viamão. Bar length encodes the attraction balance, blue for positive and red for negative.

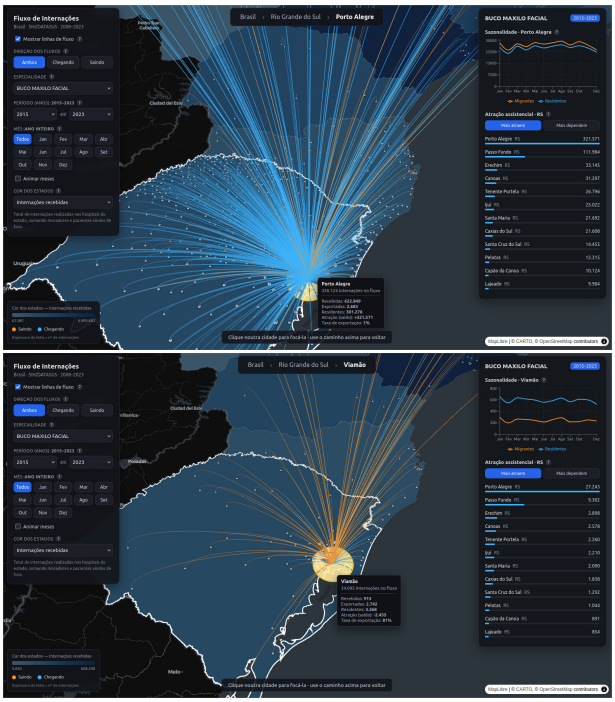


Fig. 6: City level view of two municipalities in Rio Grande do Sul for facial surgery (Buco Maxilo Facial), 2015 to 2023. Porto Alegre (top) is reached by many blue arcs, since patients arrive from across the state and from other states. Viamão (bottom) shows mostly orange arcs, since its residents leave for other cities. The two are the top attracting and top dependent municipalities in the ranking of Figure 5.

Alegre attracts far more patients than any other city, followed by Passo Fundo and Santa Maria, while Viamão, Alvorada, and Gravataí appear among the municipalities that depend on care produced elsewhere. A small number of hubs therefore sustains many surrounding towns.

The map shows the same balance for a single specialty. Figure 6 draws facial surgery over 2015 to 2023 for the two extremes of the ranking. Porto Alegre is covered by incoming blue arcs from across the state and from other states, whereas Viamão shows mostly outgoing orange arcs. The width and direction of the arcs report the export of patients for the chosen specialty.

4.3 Seasonal patterns

The third theme is seasonality, and whether a peak month falls on residents or on incoming patients. To gauge how common each pattern is, we built two seasonal profiles for every municipality with at least 50,000 admissions between 2015 and 2019, 191 in total. Each profile is a twelve-point curve of admissions by calendar month, pooled over the five years, one for resident admissions and one for incoming ones. We then took the Pearson correlation between the two profiles. Pooling by calendar month means the correlation reflects within-year seasonal shape, not any multi-year trend. The two profiles are positively correlated in 96 percent of these municipalities, and in 81 percent they reach $r \geq 0.5$, our chosen threshold for a strong relation, so co-movement is the dominant pattern. Figure 7 shows six municipalities across the twelve months of 2021 that span the range we observed. Vitória da Conquista (BA) is a co-movement case, where residents and incoming patients rise and fall together. Ceará-Mirim (RN) shows both groups climbing sharply from March, a clear seasonal peak in both series.

The other four cities show patterns that do not generalize. In Muriáe (MG) the incoming patients peak several times while residents stay flat. In Gurupi (TO) the resident and incoming peaks alternate through the year. In Luziânia (GO) and Teutônia (RS) a rise in residents coincides with a fall in incoming patients, which suggests that residents are served first when local demand grows; Teutônia also peaks in winter.

This bears on the question of overload in peak months. Where the two series move together, which is the common case measured above, a peak is not driven by a sudden influx from outside. In the minority of cities where they diverge, such as Luziânia and Teutônia, the system appears to favor residents during the busiest months. Because the behavior varies, the practical value of the monthly view is local: an institution can read the curve for its own city and prepare for its own peak.

5 Conclusion and Future Work

We presented an interactive flow map of hospital admissions in the Brazilian public system. The tool draws each admission as an arc from the patient's residence to the hospital municipality, works at the national, state, and municipal levels, and lets the user filter by specialty, year, and month. Through usage scenarios we illustrated how it can bring out gaps in care for a specialty, the hub cities that concentrate the care of a region, and the seasonal balance between resident and incoming patients. The contribution is the visualization itself, which is built to expose movement that the official tabular access leaves implicit.

The central limitation is that we have not evaluated the tool with users. The scenarios in Section 4 are hand-picked illustrations, so they show that the tool can produce these readings, not that it helps an analyst reach them faster or more reliably than the existing portal. We therefore make no claim of measured effectiveness, and a study with health managers, outlined below, is the natural next step.

A second limitation concerns the specialty field. SIH/SUS records the specialty of the hospital bed (ESPEC), as noted in Section 3, and this field is sparsely filled. A few categories, mainly oral and maxillofacial surgery, general surgery, cardiology, and a combined surgical and diagnostic category, account for almost all admissions, while specialties such as oncology and endocrinology appear in very few records, and several others had too little data to keep. Much of this care is likely registered through the procedure and diagnosis fields rather than the bed specialty, so a filter built on this single field gives only a partial view.



Fig. 7: Monthly resident and incoming admissions for six municipalities in 2021. The shape of the curve changes from city to city, from residents and incoming patients moving together (Vitória da Conquista, BA) to shared peaks (Ceará-Mirim, RN), repeated incoming peaks (Muriáe, MG), alternating peaks (Gurupi, TO), and resident peaks that coincide with drops in incoming patients (Luziânia, GO and Teutônia, RS).

Other limits follow from the same design choices. The latest year is incomplete in its final months, so the scenarios rely on complete years. State-level arcs start from state centroids, and the few municipalities without coordinates are left out. The per-capita metric depends on IBGE population figures and so covers only 2008 to 2021. Finally, the tool shows where demand concentrates but does not measure hospital capacity, so it points to possible overload rather than confirming it.

Future work follows directly from these limits. The first priority is the evaluation that this paper lacks. We plan two lightweight studies: a walkthrough in which a small group of health-data analysts carry out defined tasks with the tool, and a comparison against the DATASUS portal that measures the time to answer a handful of representative questions. Both are proposed here rather than reported, since we have not yet run them.

The remaining directions address the data limits. Reading the procedure and diagnosis fields would recover the specialties that the bed-specialty field misses and would widen the analysis of care gaps. Joining the flows to the number of SUS beds would turn the signal of concentration into a measure of load. Finer geography, such as health regions and population-weighted centroids, would sharpen the map, and the cost and length-of-stay fields would add a financial and outcome dimension to the movement of patients.

The SIH/SUS data used here is public and distributed by DATASUS. The source code of the tool is available from the authors on request.

References

[1] I. Boyandin, E. Bertini, P. Bak, and D. Lalanne. Flowstrates: An approach for visual exploration of temporal origin-destination data. *Comput. Graph. Forum*, 30:971–980, 06 2011. doi: 10.1111/j.1467-8659.2011.01946.x 1

[2] A. Graser, J. Schmidt, F. Roth, and N. Brändle. Untangling origin-destination flows in geographic information systems. *Information Visualization*, 18:147387161773812, 11 2017. doi: 10.1177/1473871617738122 1

[3] B. Jenny, D. Stephen, I. Muehlenhaus, B. Marston, R. Sharma, E. Zhang et al. Design principles for origin-destination flow maps. *Cartography and Geographic Information Science*, 45, 12 2016. doi: 10.1080/15230406.2016.1262280 1

[4] E. X. G. d. Oliveira, M. S. Carvalho, and C. Travassos. Territórios do sistema Único de saúde: mapeamento das redes de atenção hospitalar. *Cadernos de Saúde Pública*, 20(2):386–402, Mar 2004. doi: 10.1590/S0102-311X2004000200006 1

[5] J. S. Y. Rocha, R. A. Monteiro, and M. L. a. Moreira. Hospitalization flow in the public and private systems in the state of sao paulo, brazil. *Revista de Saúde Pública*, 49, 2015. doi: 10.1590/S0034-8910.2015049005696 1

[6] B. Shneiderman. The eyes have it: A task by data type taxonomy for information visualizations. In *Proceedings of the 1996 IEEE Symposium on Visual Languages*, VL '96, p. 336. IEEE Computer Society, USA, 1996. 1, 2

[7] J. Wood, J. Dykes, and A. Slingsby. Visualisation of origins, destinations and flows with od maps. *Cartographic Journal, The*, 47:117–129, 05 2010. doi: 10.1179/000870410X12658023467367 1

[8] D. R. Xavier, R. A. D. d. Oliveira, C. Barcellos, R. d. F. Saldanha, W. M. Ramalho, J. Laguardia et al. As regiões de saúde no brasil segundo internações: método para apoio na regionalização de saúde. *Cadernos de Saúde Pública*, 35:e00076118, 2019. doi: 10.1590/0102-311X00076118 1

[9] J. S. Yi, Y. a. Kang, J. Stasko, and J. Jacko. Toward a deeper understanding of the role of interaction in information visualization. *IEEE Transactions on Visualization and Computer Graphics*, 13(6):1224–1231, Nov. 2007. doi: 10.1109/TVCG.2007.70515 1